

1	Course Name:
	Advanced Routing
2	Course Code:
	INB35605
3	Year and Semester Offered:
	Year 3 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	INB35504 Network Interconnections
6	Course Synopsis:
	This course teaches students how to implement, monitor, and maintain routing services in an enterprise network. The Students will learn how to plan, configure, and verify the implementation of complex enterprise LAN and WAN routing solutions, using a range of routing protocols in IPv4 and IPv6 environments. The routing protocols covers in the course are Enhanced Interior Gateway Routing Protocol (EIGRP), Open Shortest Path First (OSPF) and Border Gateway Protocol (BGP). The students will also learn on how to manipulate and controlling the network traffics by implementing route redistribution and path control. This course give a comprehensive labs emphasize hands-on learning and practice to reinforce configuration skills.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Determine appropriate techniques related to advanced routing implementation in an Enterprise Network. 2. Performs the implementation of advanced routing configurations in an Enterprise Network. 3. Identifies network issues and advanced routing techniques in an Enterprise Network.
8	Content Outline:
	Topic
	1. DHCP to across multiple LANs
	2. EIGRP for IPv4 in a small to medium-sized business network
	3. Advanced EIGRP features
	4. Multiarea OSPF for IPv4 to enable internetwork communications
	5. Multiarea OSPFv3
	6. BGP
	7. BGP advanced features
	8. BGP path selection

	9. Route redistribution
	10. The impact of VFR and MPLS on routing decisions
	11. Implement DMVPN tunnels
	12. IPsec DMVPN with Pre-Shared Authentication